
Chapter 7

Noise

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Outline

- **Statistical Characteristics of Noise**
- **Types of Noise**
- **Representation of Noise in Circuits**
- **Noise in Single-Stage Amplifiers**
- **Noise in Current Mirrors**
- **Noise in Differential Pairs**
- **Noise-Power Trade-Off**
- **Noise Bandwidth**
- **Problem of Input Noise Integration**



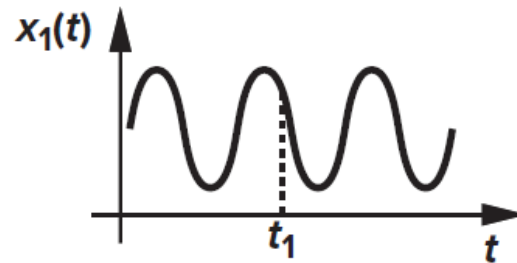
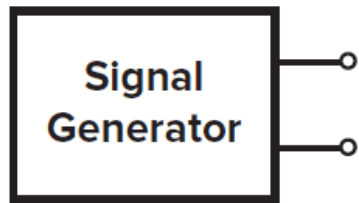
Outline

- **Statistical Characteristics of Noise**
 - Noise spectrum
 - Amplitude distribution
 - Correlated and uncorrelated noise sources
 - Signal-to-noise ratio
 - Procedure of noise analysis

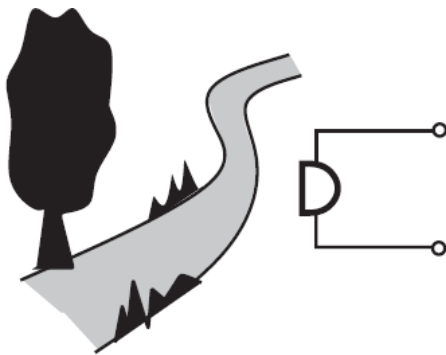


What is noise?

- General definition: **unwanted signal** in a system
 - **Interference (deterministic)** and **noise (random)**
- More accurate definition: **unwanted random signal**



Predictable

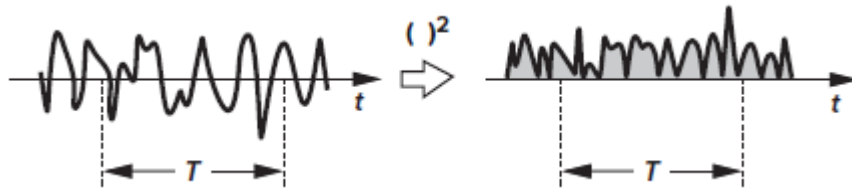


Random



Noise Power

Noise amplitude is unpredictable, how to deal with noise?
Fortunately, the average power is often predictable.

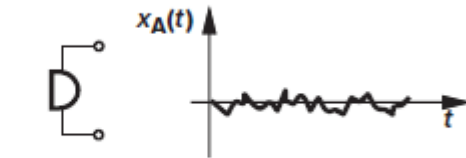


$$P_{av} = \frac{1}{T} \int_{-T/2}^{+T/2} \frac{v^2(t)}{R_L} dt$$

$$P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} \frac{x^2(t)}{R_L} dt$$

$$P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} x^2(t) dt$$

Commonly used definition
unit is V²!



Small power

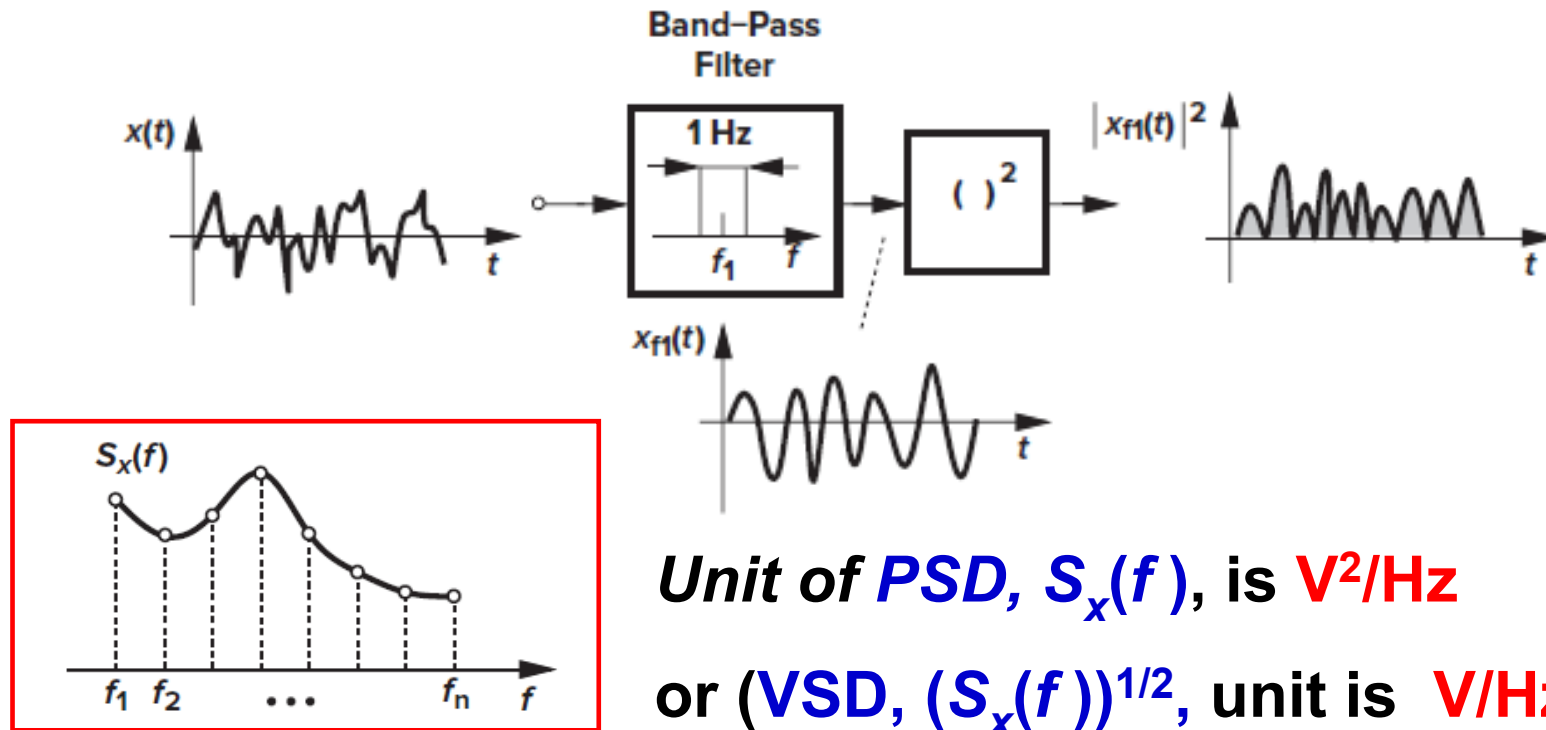


Large power



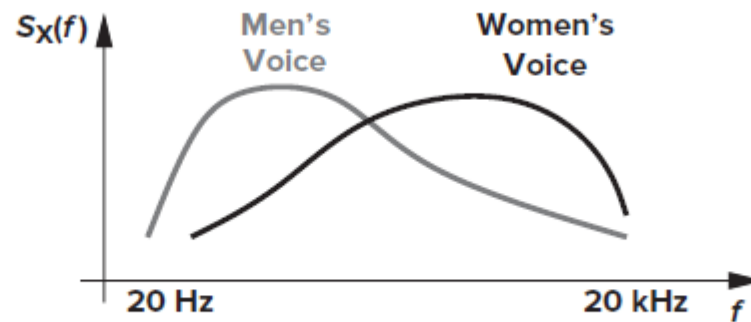
Noise Spectrum and PSD

- **Definition:** Noise power carried by different frequency component, or **Power Spectrum Density (PSD)**
- Use a 1-Hz moving window (bandpass filter) to measure

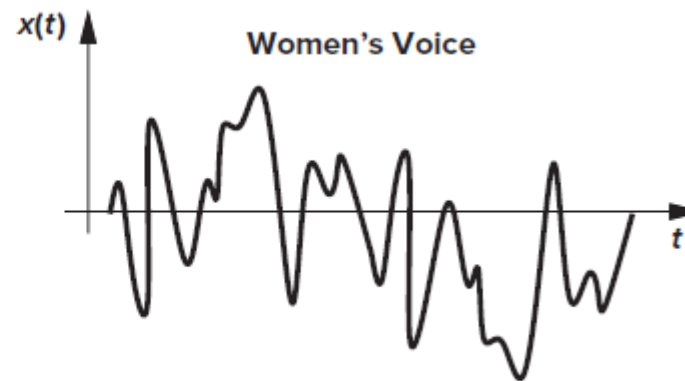
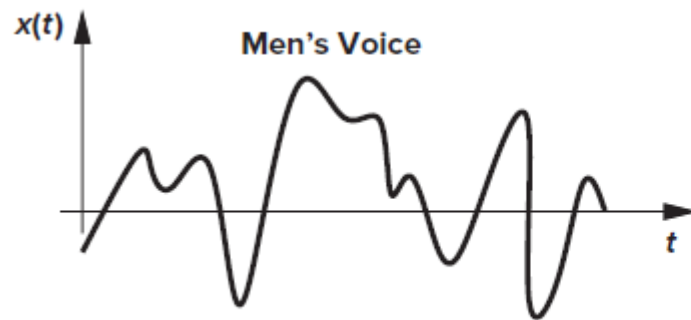


Men and Women's Voice Spectrum

- PSD analysis is a useful method to analyze random signals (or noise)



(a)



(b)

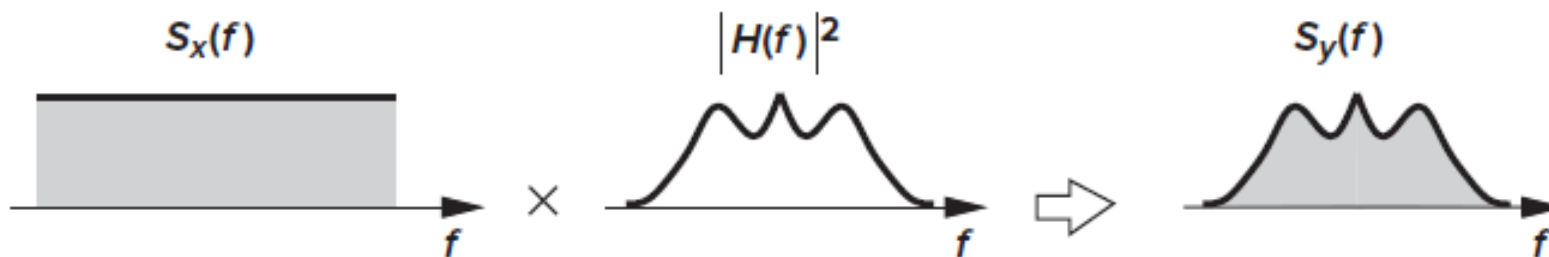


Noise Shaping of LTI systems

- **Theorem:** A signal with PSD $S_x(f)$ passes a LTI system with transfer function of $H(f)$, the output PSD is

$$S_Y(f) = S_x(f)|H(f)|^2$$

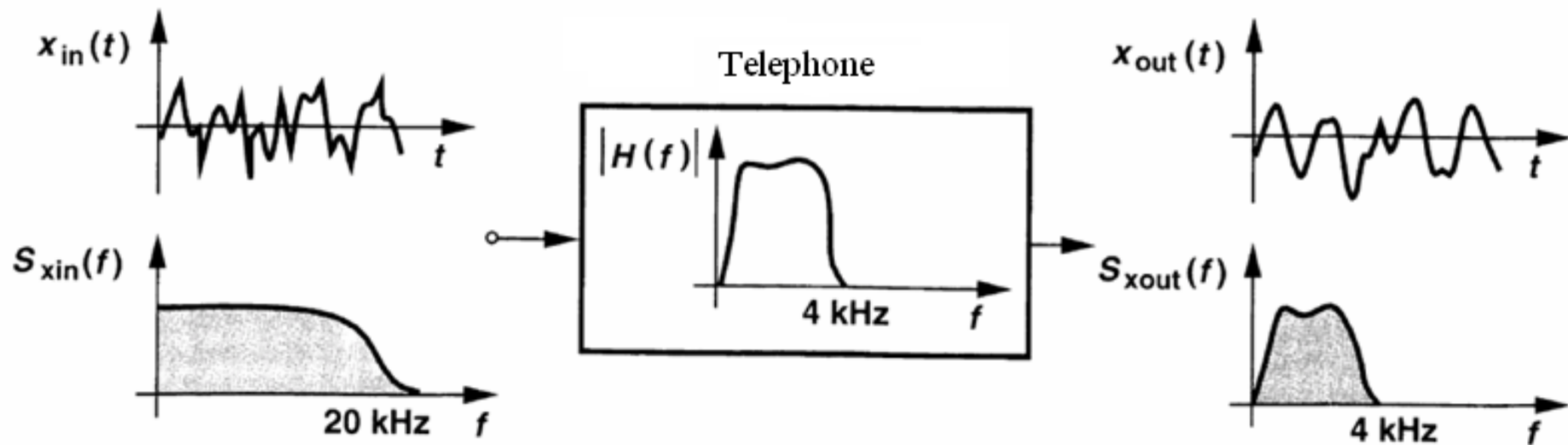
- White spectrum (**white noise**): a common type of noise
 - PSD is constant at all frequencies (is this practical?)
 - More general definition: noise spectrum is flat in the band of interest
- White spectrum passes a LTI system, output is a “**shaped spectrum**”



Noise Shaping Example

Voice passes a narrowband telephone channel

– Why voice is more difficult to recognize?



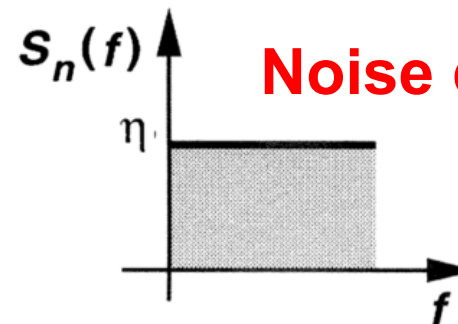
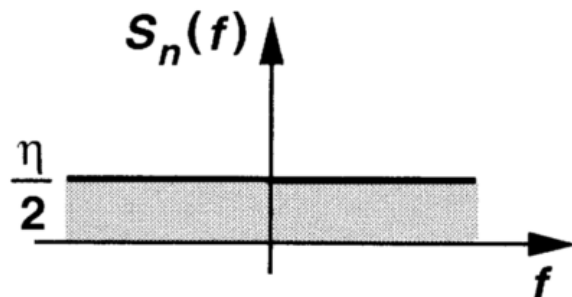
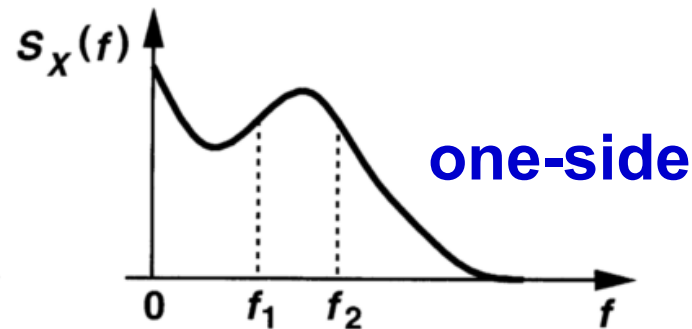
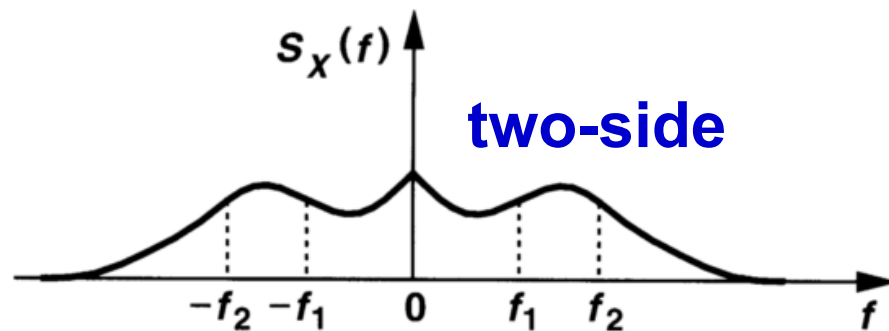
– High-frequency component is suppressed



Two-sided and one-sided spectra

Real signal's spectrum $x(f)$ is an even function of f

Can be expressed as either **two-side** or **one-side** spectrum

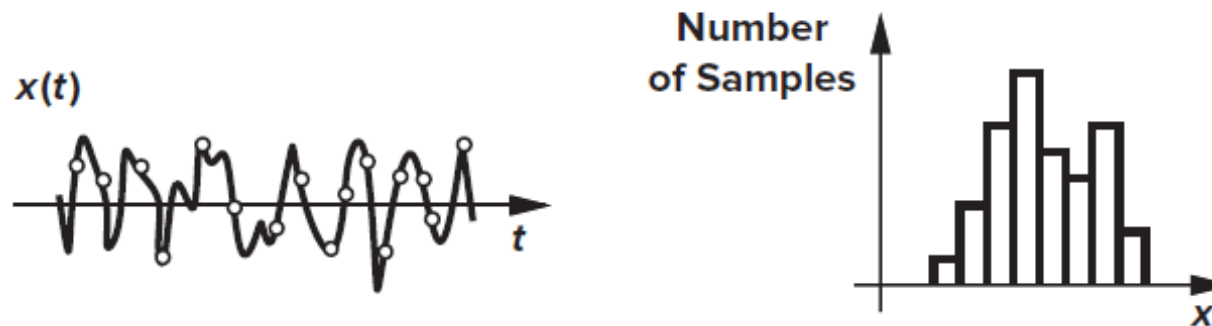


Noise density doubled
Try to be strict!!



Amplitude Distribution of Noise

- Another way to represent noise: **Statistical method**
- Use large number of amplitude samples to obtain the **probability distribution function (PDF)**



- An important PDF: **Gaussian (or normal) distribution**
- Many natural phenomenon exhibit Gaussian PDF, why?

$$p_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \frac{-(x - m)^2}{2\sigma^2}$$

What do m and σ stand for?
 σ^2 is the average noise power

Effect of two or more noise sources

- When several noise sources add up, how much is the overall noise power?

$$\begin{aligned} P_{av} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} [x_1(t) + x_2(t)]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} x_1^2(t) dt + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} x_2^2(t) dt \\ &\quad + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} 2x_1(t)x_2(t) dt \\ &= P_{av1} + P_{av2} + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} 2x_1(t)x_2(t) dt \end{aligned}$$



Correlated and Uncorrelated Noise Sources

- **The concept of correlation: how similar the signals are**

$$P_{av} = P_{av1} + P_{av2} + \boxed{\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} 2x_1(t)x_2(t)dt}$$

Correlation of $x_1(t)$ and $x_2(t)$

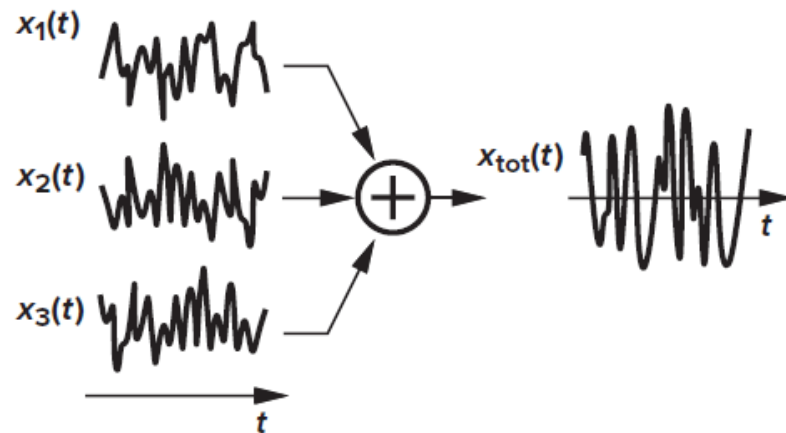
- **Uncorrelated noise: generated from independent devices. The correlation is zero, $P_{av} = P_{av1} + P_{av2}$, power adds up**
- **Correlated noise: noise sources are dependent on each other to some extent. The correlation is nonzero. Noise can be enhanced or cancelled.**



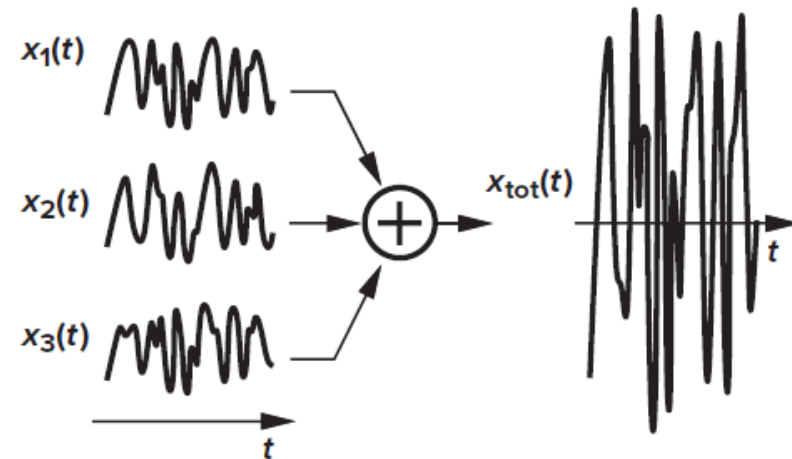
An example: stadium

- Before the game starts, **uncorrelated noises, power adds up**
- During the games, people scream) simultaneously, producing **correlated noise with much higher power**

Uncorrelated



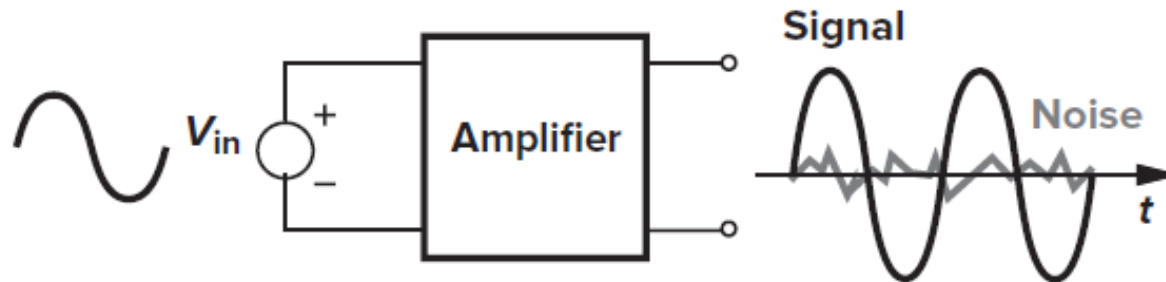
Correlated



For most cases: noise sources are uncorrelated.

Signal to noise ratio (SNR)

- When a signal passes an amplifier, the output contains both the signal and the noise. The signal power must be much larger than the noise power
- **SNR** is ratio of signal power over noise power



$$\text{SNR} = \frac{P_{sig}}{P_{noise}}$$

$$P_{noise} = \int_{-\infty}^{+\infty} S_{noise}(f) df$$

$$P_{sig} = ?$$

- **Can an amplifier increase the SNR?**

Procedure to analyze noise

1. Identify noise sources, analyze their spectra
2. Find the **transfer function from each noise to the output node.**
3. Calculate noise spectrum contributed by each noise source
4. Add all spectra, **pay attention to correlated and uncorrelated noise sources.**



Outline

- Statistical Characteristics of Noise
- **Types of Noise**
 - Resistor thermal noise
 - MOSFET channel thermal noise
 - MOSFET Ohmic (parasitic resistor) noise
 - MOSFET flicker noise



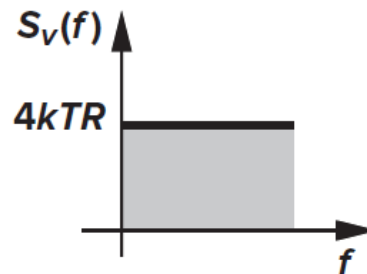
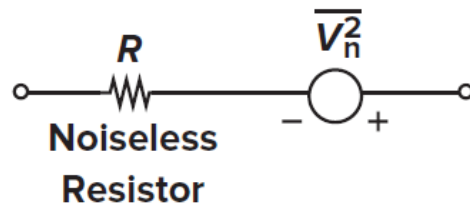
Types of noise

- Two types of noise can corrupt analog signals
 - Device noise
 - Environmental noise
 - Example: noise from Supply or Ground lines
- This chapter focuses on device noise



Resistor Thermal Noise

- The random motion of electron introduces fluctuation in the voltage across the conductor
 - Thermal noise
 - Dependent on temperature
- Noise modeling
 - A noise voltage source in series
 - With one-side spectrum



$$S_v(f) = 4kTR$$

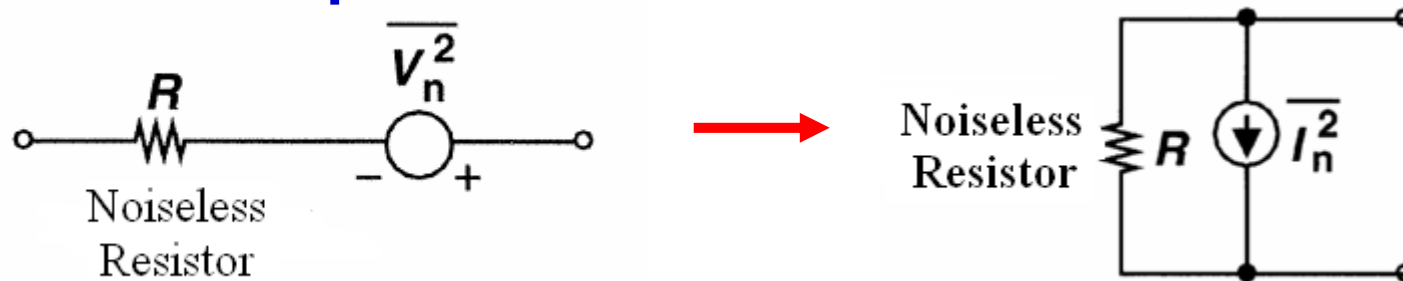
$$\text{or: } \overline{V_n^2} = 4kTR$$

$$\text{for } f \geq 0 \quad \text{unit: } V^2/\text{Hz}$$

Resistor Thermal Noise

$$\overline{V_n^2} = 4kTR$$

- $k = 1.38 \times 10^{-23}$ J/K (Boltzman constant),
- T : Absolute temperature
- 50Ω resistor at 300K $\rightarrow \overline{V_n^2} = 8.28 \times 10^{-19}$ V²/Hz
- Resistor noise can also be modeled as a current source in parallel

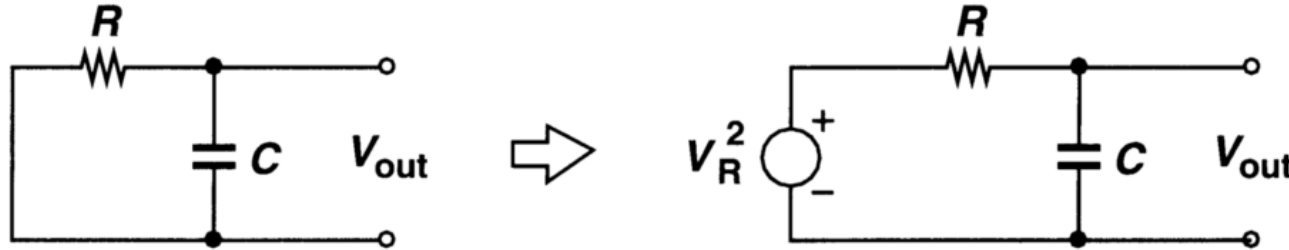


$$\overline{V_n^2} = 4kTR \Rightarrow \overline{I_n^2} = \frac{4kT}{R} \quad (\text{unit: A}^2/\text{Hz})$$



Noise Calculation

Example: Noise generated in a low-pass filter



$$\frac{V_{out}}{V_R}(s) = \frac{1}{RCs + 1} \quad S_{out}(f) = S_R(f) \left| \frac{V_{out}}{V_R}(j\omega) \right|^2 = 4kTR \frac{1}{4\pi^2 R^2 C^2 f^2 + 1}$$

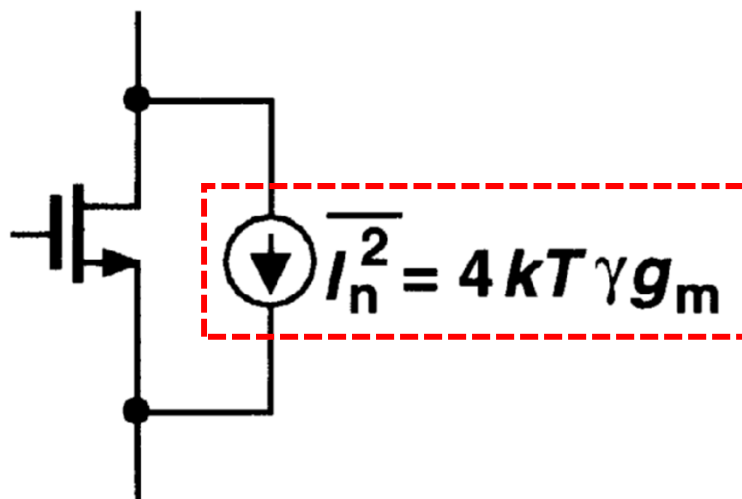
$$P_{n,out} = \int_0^\infty \frac{4kTR}{4\pi^2 R^2 C^2 f^2 + 1} df = \frac{2kT}{\pi C} \tan^{-1} u \Big|_{u=0}^{u=\infty} \Rightarrow P_{n,out} = \frac{kT}{C}$$

- **Very interesting and important conclusion:**
R generates noise but noise power is determined by C!
Why?



MOSFET Channel Thermal Noise

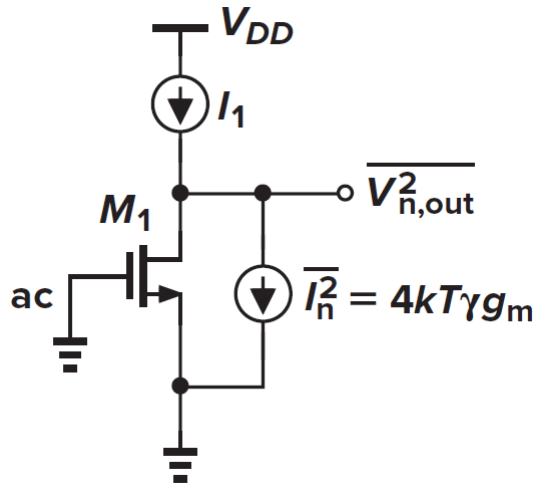
- For long-channel MOS in saturation, channel noise is modeled as a current source in parallel



Long channel: $\gamma \approx 2/3$,
Submicron devices, γ increases, up to 2.5



Example: output noise calculation



- Common source with ideal current source load
- Input AC grounded, to calculate the output noise

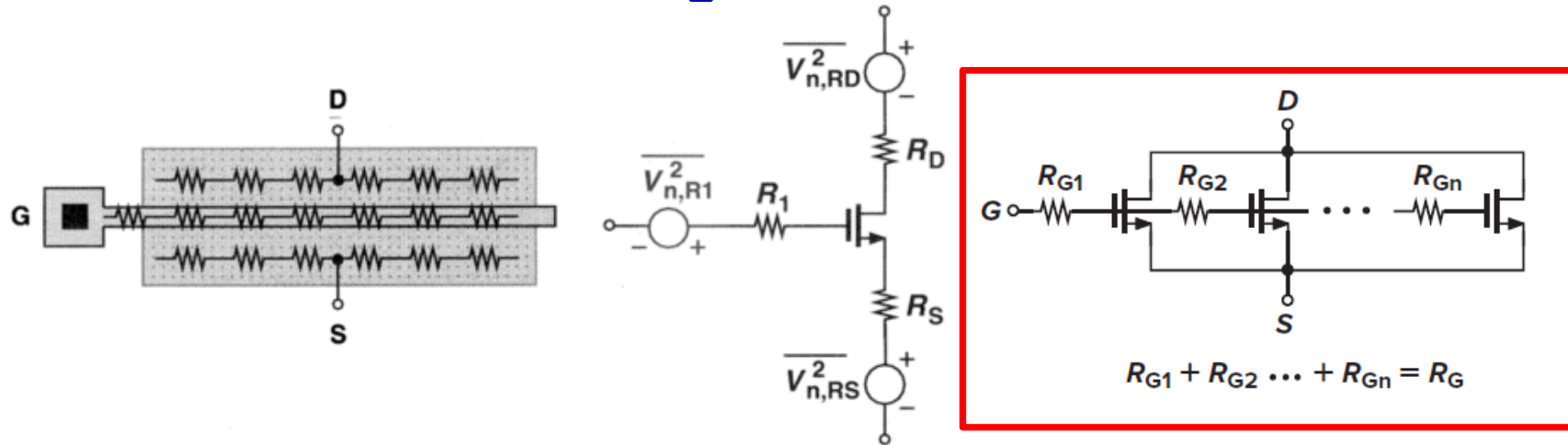
$$\begin{aligned}\overline{V_n^2} &= \overline{I_n^2} r_O^2 \\ &= (4kT\gamma g_m) r_O^2\end{aligned}$$

- Important observation:
 - $g_m \downarrow \rightarrow$ noise power $\downarrow$
 - Output noise is independent of where the input is
 - r_o does not produces noise, why?



MOSFET Ohmic Thermal Noise

Parasitic resistors also generate thermal noise



For wide MOS, drain/source resistance is negligible

Should pay attention to the gate resistance

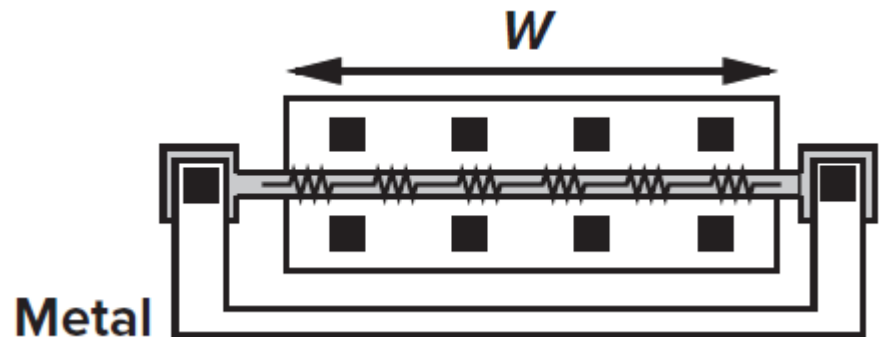
Total gate resistance noise power

$$\overline{V_{nRG}^2} = 4kT R_G / 3 \quad \text{Why?}$$



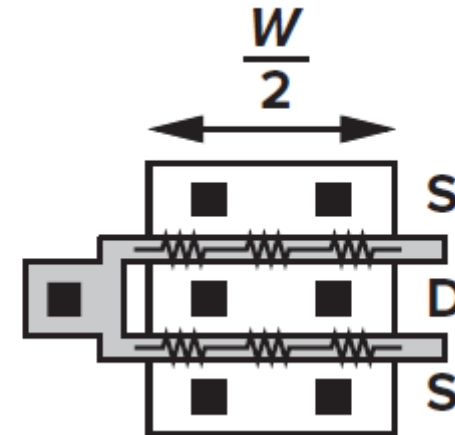
Good layout to reduce gate noise

- Adding contacts to both sides



$$R_G \rightarrow R_G/4$$

- Folding (multi-fingers)

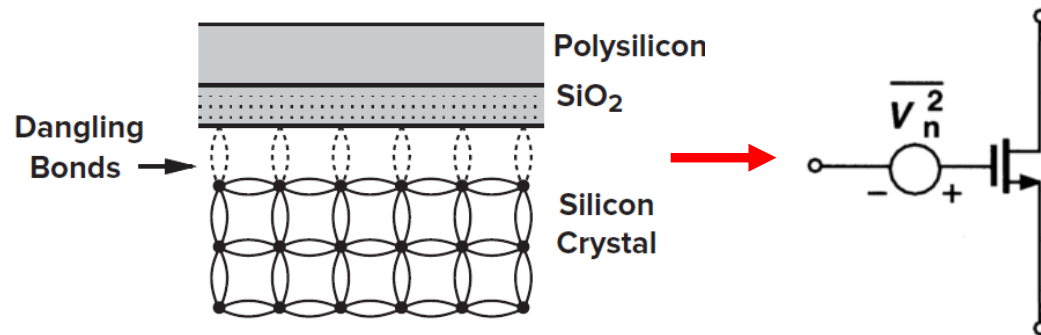


$$R_G \rightarrow R_G/4$$



MOSFET Flicker Noise

- Dangling Bonds at the surface → Extra energy states
→ carriers are trapped randomly and released later
– → **Flicker noise**



$$\overline{V_n^2} = \frac{K}{C_{ox}WL} \frac{1}{f}$$

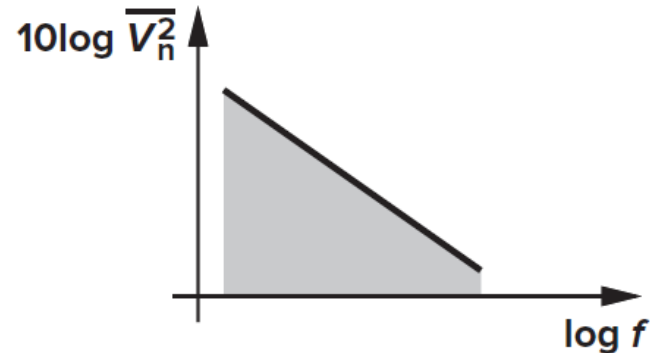
***K* is a process-dependent constant (unit: V²F)**

- More difficult to predict, depending on “cleanness”
- Modeled as a voltage source in series with the gate



Frequency Dependence of Flicker Noise

$$\overline{V_n^2} = \frac{K}{C_{ox} WL} \frac{1}{f}$$

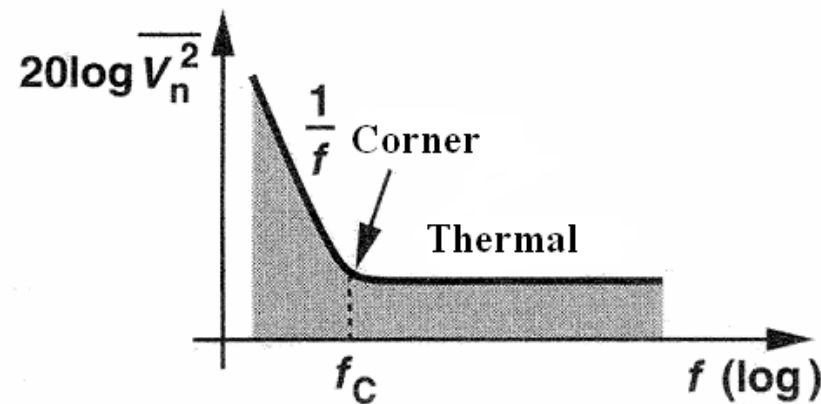


- PSD of flicker noise is proportional to $1/f$, why?
- Also called **1/f noise**, does not depend on T and g_m
- PMOS is better than NMOS, why?
- MOS 1/f noise is more serious than BJT, why?
- How to reduce 1/f noise?
 - Large-size transistors
- An interesting question: noise power is infinite at DC.
 - Is this true?



Flicker Noise Corner Frequency

- The frequency at which the PSD of flicker noise equals to the PSD of thermal noise



$$4kTg_m\gamma \approx \frac{K}{C_{ox}WL} \frac{1}{f_c} g_m^2$$

$$f_c \approx \frac{K}{C_{ox}WL} \frac{g_m}{4\gamma kT}$$

For given L, the corner frequency is relatively constant, why?



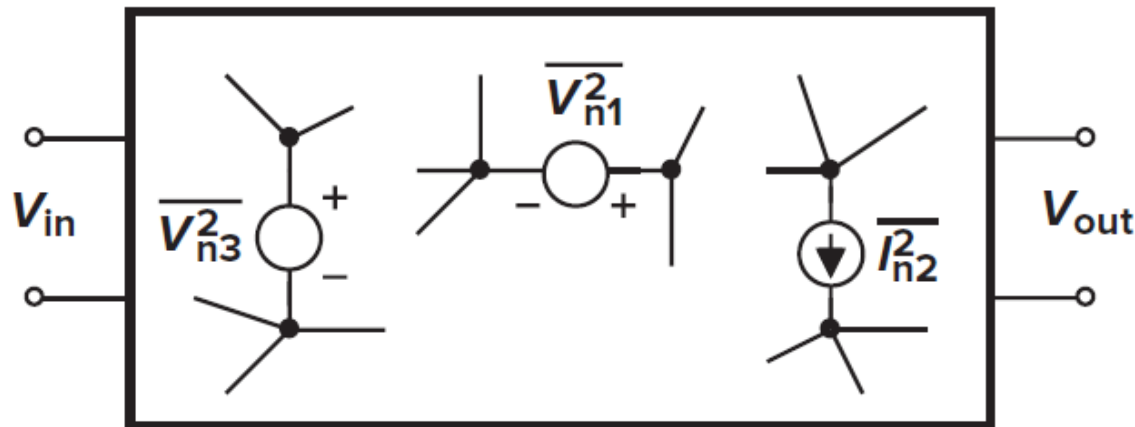
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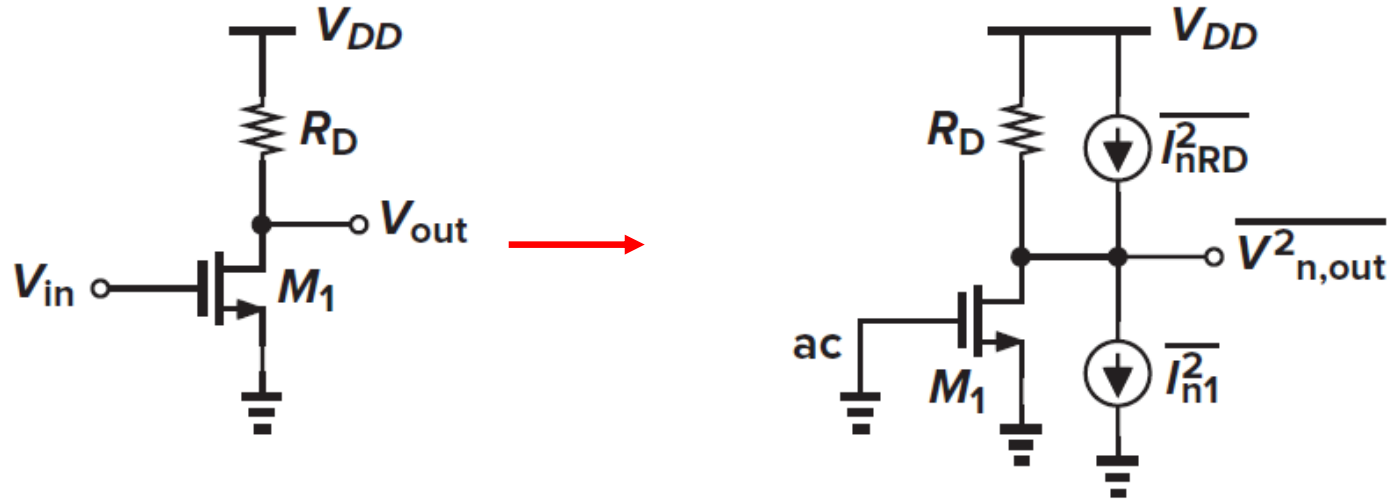


Representation of Noise in Circuits

- Many circuits can be modeled as a two-port network Including various noise sources
- How to quantify the effect of each noise?
- Input grounded (AC), calculate the total output noise.



Example 7.7: Calculate total output noise



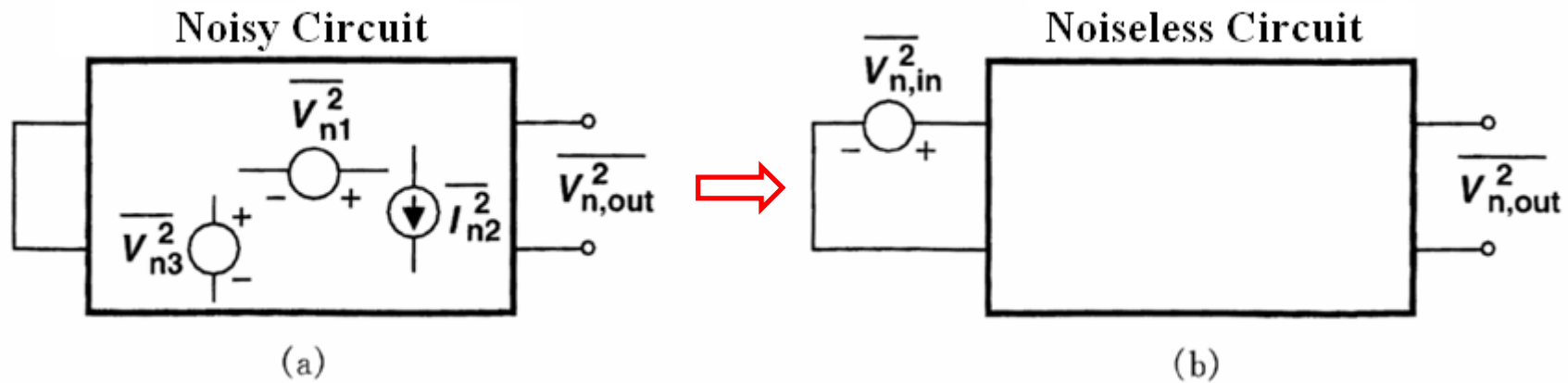
Total noise PSD = M1 thermal + M1 flicker + R_D thermal

$$\overline{V_{n,out}^2} = \left(4kT\gamma g_m + \frac{K}{C_{ox}WL} \frac{1}{f} g_m^2 + \frac{4kT}{R_D} \right) R_D^2$$



Input Referred Noise

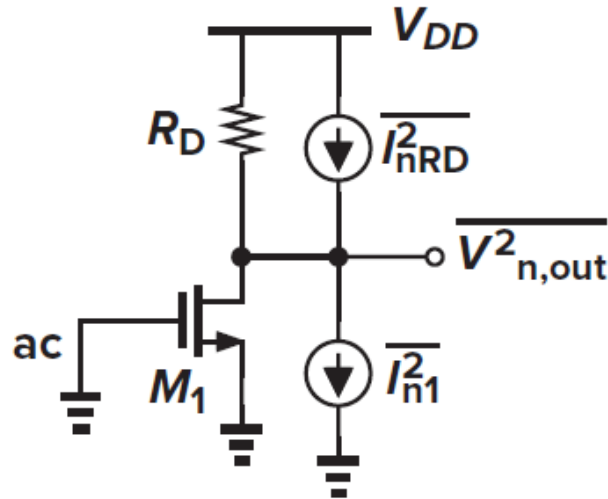
- Convenient to show how small an input signal the circuit can detect with a required SNR
- Input referred noise is fictitious quantity that allows comparison between different circuits.
- Can't be measured at the input



$$\overline{V_{n,out}^2} = A_v^2 \overline{V_{n,in}^2}$$



Example 7.7 revisited



- Calculate the total input-referred noise

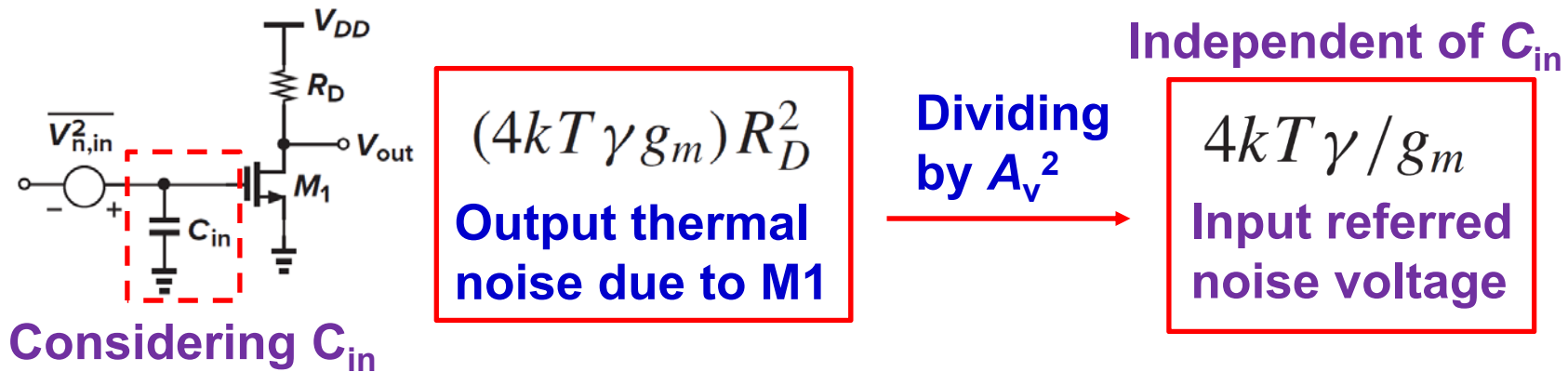
$$\overline{V_{n,in}^2} = \frac{\overline{V_{n,out}^2}}{A_v^2} = \frac{\overline{V_{n,out}^2}}{g_m^2 R_D^2}$$

$$\begin{aligned} \overline{V_{n,in}^2} &= \left(4kT\gamma g_m + \frac{K}{C_{ox}WL} \frac{1}{f} g_m^2 + \frac{4kT}{R_D} \right) R_D^2 \frac{1}{g_m^2 R_D^2} \\ &= 4kT \frac{\gamma}{g_m} + \frac{K}{C_{ox}WL} \frac{1}{f} + \frac{4kT}{g_m^2 R_D} \end{aligned}$$

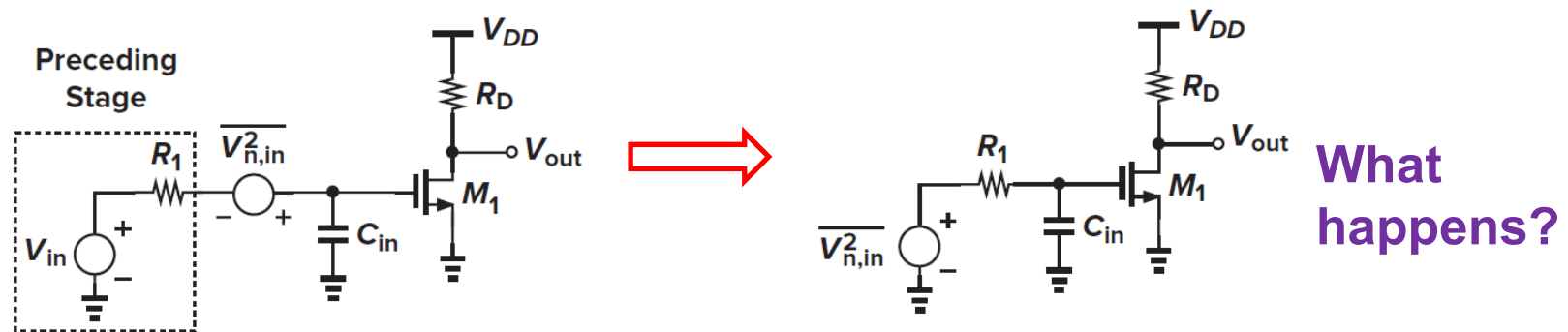


Considering the driving stage

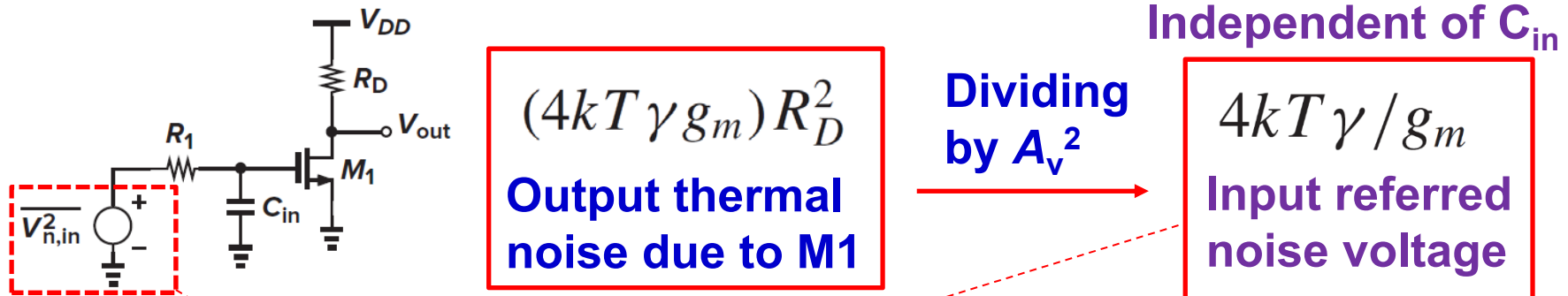
- Using a single voltage source to represent the input-referred noise is usually insufficient.



- Considering the driving stage with resistance of R_1



Considering the driving stage



$$\overline{V_{n,out}^2} = \overline{V_{n,in}^2} \left| \frac{1}{R_1 C_{in} j\omega + 1} \right|^2 (g_m R_D)^2$$

$$= \frac{4kT\gamma g_m R_D^2}{R_1^2 C_{in}^2 \omega^2 + 1}$$

Seems that the output noise can be suppressed with a large R_1

Incorrect!

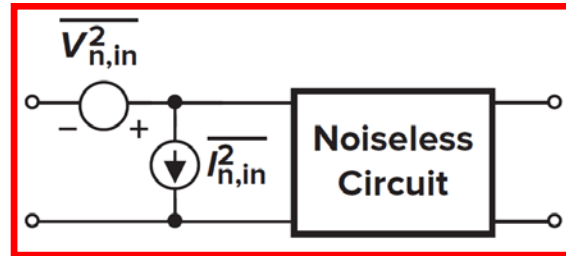
$$\Rightarrow \overline{V_{n,out}^2} \neq (4kT\gamma g_m)R_D^2$$

Why?

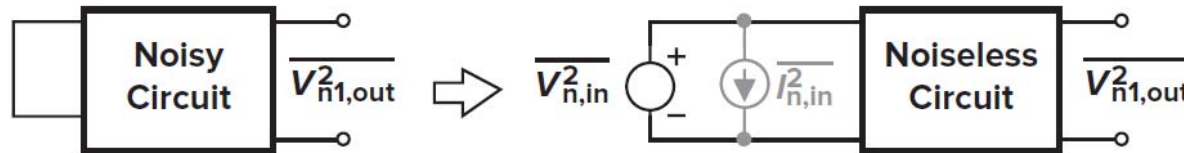


Complete input network

- If the input impedance is finite (Eg.: $1/(j\omega C_1)$), and the source impedance is large, the input-referred noise voltage source is incorrectly suppressed
- Solution: adding a parallel current source

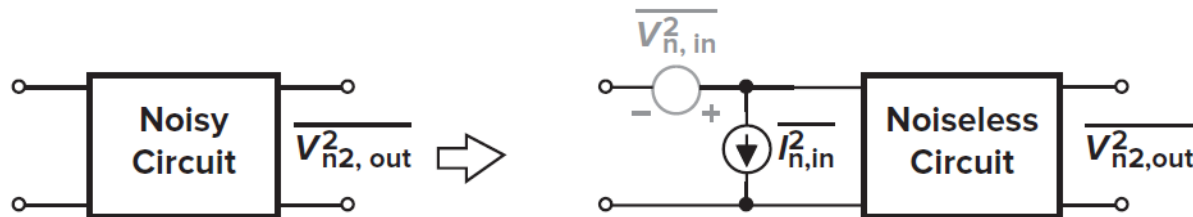


$R1 = 0$



Voltage
source
takes effect

$R1 = \infty$

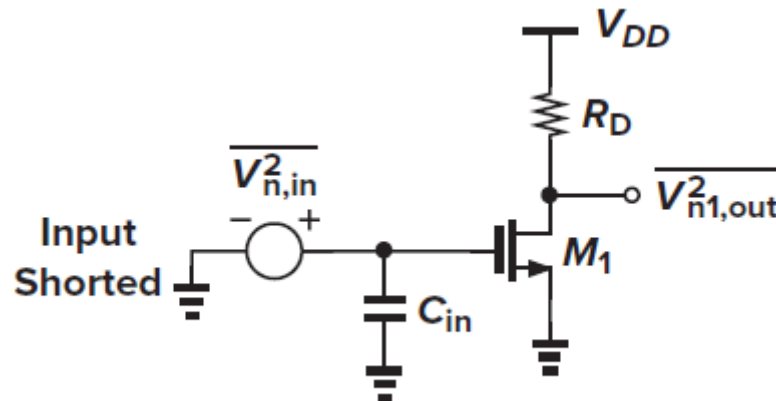


Current
source
takes effect

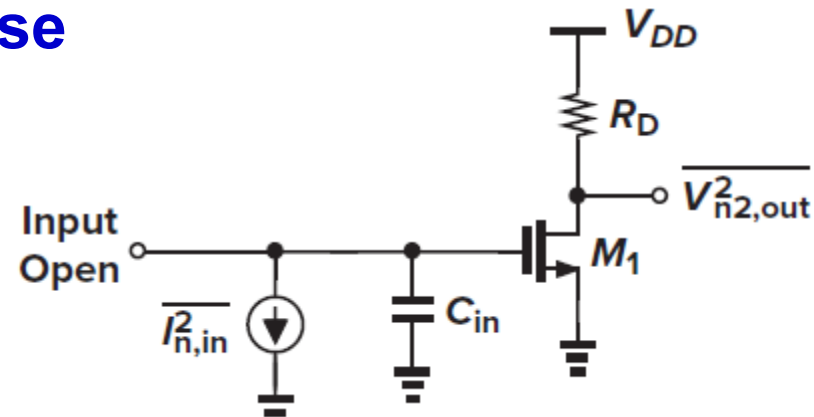


Example: Calculate $V_{n,in}$ and $I_{n,in}$

Considering only thermal noise



$$\overline{V_{n,in}^2} = 4kT \frac{\gamma}{g_m} + \frac{4kT}{g_m^2 R_D}$$



$$\overline{V_{n2,out}^2} = \overline{I_{n,in}^2} \left(\frac{1}{C_{in}\omega} \right)^2 g_m^2 R_D^2$$

$$\overline{V_{n2,out}^2} = \left(4kT \gamma g_m + \frac{4kT}{R_D} \right) R_D^2$$

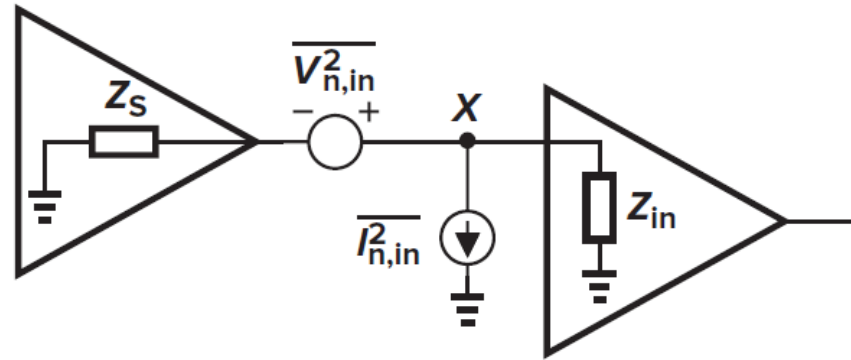
Also



$$\overline{I_{n,in}^2} = (C_{in}\omega)^2 \frac{4kT}{g_m^2} \left(\gamma g_m + \frac{1}{R_D} \right)$$



Relationship of $V_{n,in}$, $I_{n,in}$, Z_S , and Z_{in}



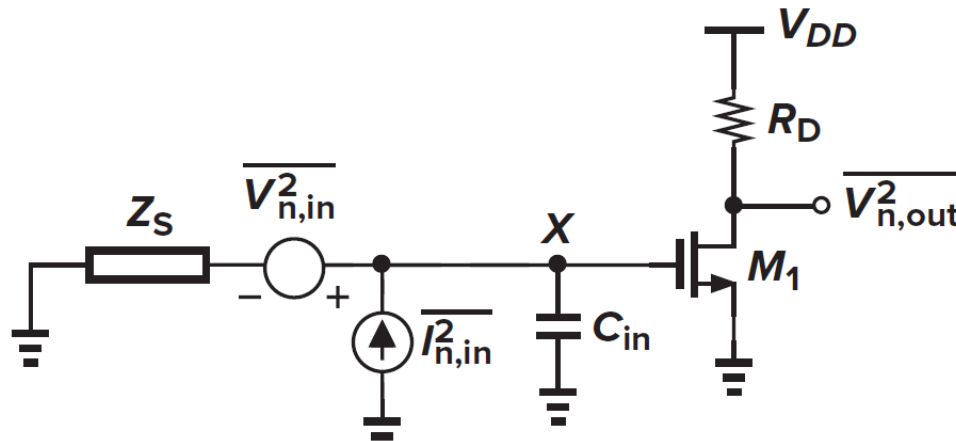
$$V_{n,X} = \frac{Z_{in}}{Z_{in} + Z_S} V_{n,in} + \frac{Z_{in} Z_S}{Z_{in} + Z_S} I_{n,in}$$

- If $\overline{I_{n,in}^2} |Z_S|^2 \ll \overline{V_{n,in}^2}$ the effect of $I_{n,in}$ can be neglected
- Z_S determines the significance of $I_{n,in}$



Do we count the noise twice?

- Although we use both $V_{n,in}$ and $I_{n,in}$, they may be correlated.



$$\overline{V_{n,in}^2} = 4kT \frac{\gamma}{g_m} + \frac{4kT}{g_m^2 R_D}$$

$$\overline{I_{n,in}^2} = (C_{in}\omega)^2 \frac{4kT}{g_m^2} \left(\gamma g_m + \frac{1}{R_D} \right)$$

$$\begin{aligned} V_{n,X} &= V_{n,in} \frac{\frac{1}{C_{in}s}}{\frac{1}{C_{in}s} + Z_S} + I_{n,in} \frac{\frac{Z_S}{C_{in}s}}{\frac{1}{C_{in}s} + Z_S} \\ &= \frac{V_{n,in} + I_{n,in} Z_S}{Z_S C_{in}s + 1} \end{aligned}$$



$$\begin{aligned} \overline{V_{n,out}^2} &= g_m^2 R_D^2 \overline{V_{n,X}^2} \\ &= 4kT \left(\gamma g_m + \frac{1}{R_D} \right) R_D^2 \end{aligned}$$

The model is correct!

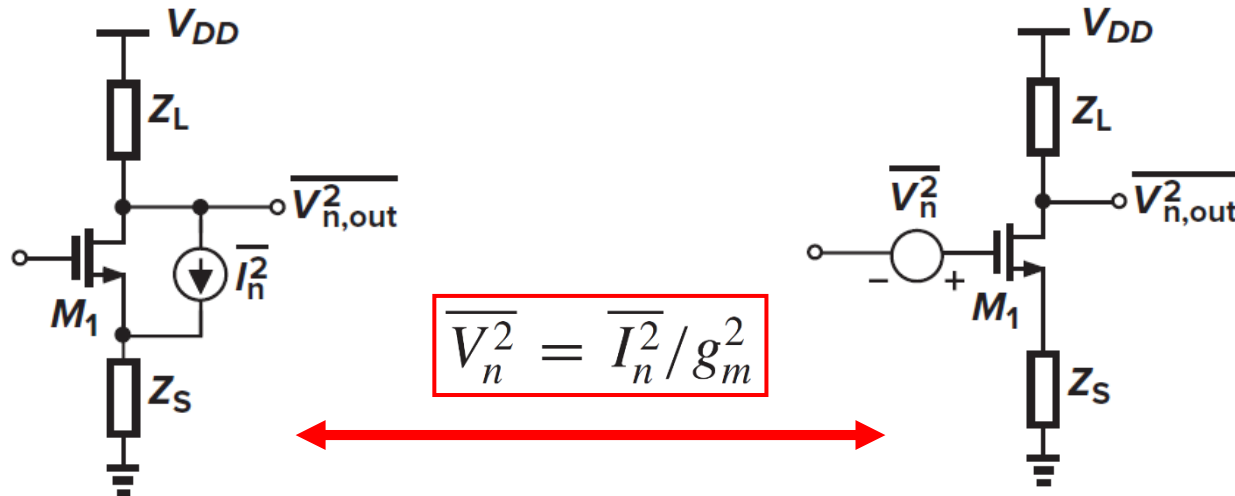


Outline

- Statistical Characteristics of Noise
- Types of Noise
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- **Noise in Single-Stage Amplifiers**
- Noise in Current Mirrors
- Noise in Differential Pairs
- Noise-Power Trade-Off
- Noise Bandwidth
- Problem of Input Noise Integration



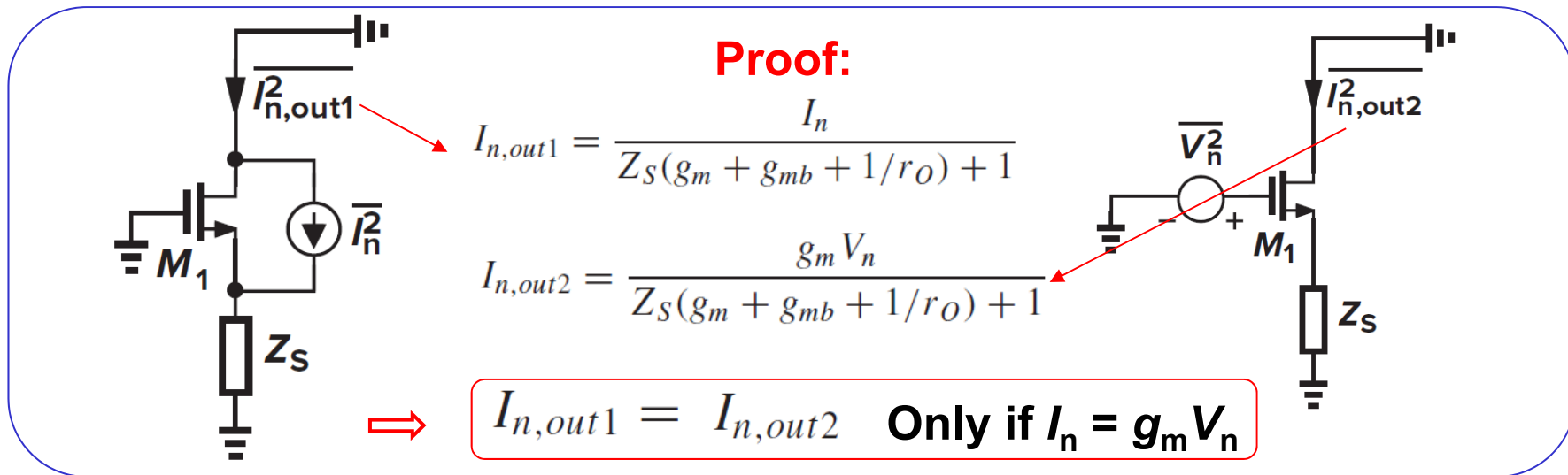
An useful lemma



$$\overline{V_n^2} = \overline{I_n^2} / g_m^2$$

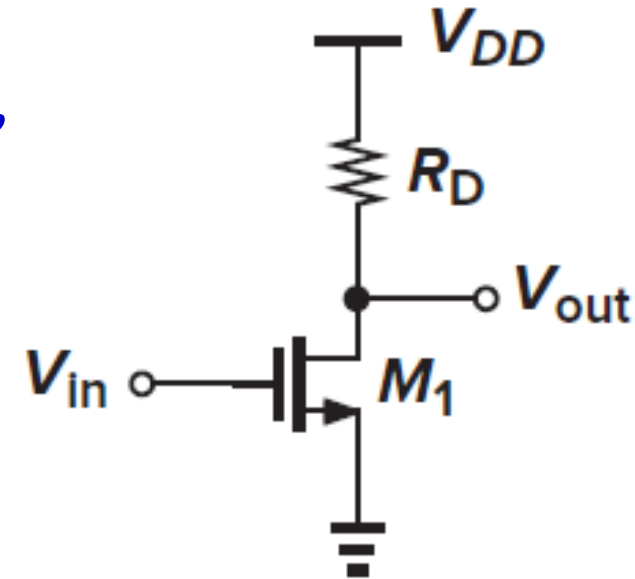
Conditions:

- Driven by a finite impedance.
- For arbitrary Z_s .
- Equivalent at low frequencies.



Common-Source Stage

- The input impedance is very large, especially at low frequency.
 - $I_{n,in}$ of CS stage can often be neglected



Input referred noise PSD:

$$\overline{V_{n,in}^2} = 4kT \left(\frac{\gamma}{g_m} + \frac{1}{g_m^2 R_D} \right) + \frac{K}{C_{ox} W L} \frac{1}{f}$$

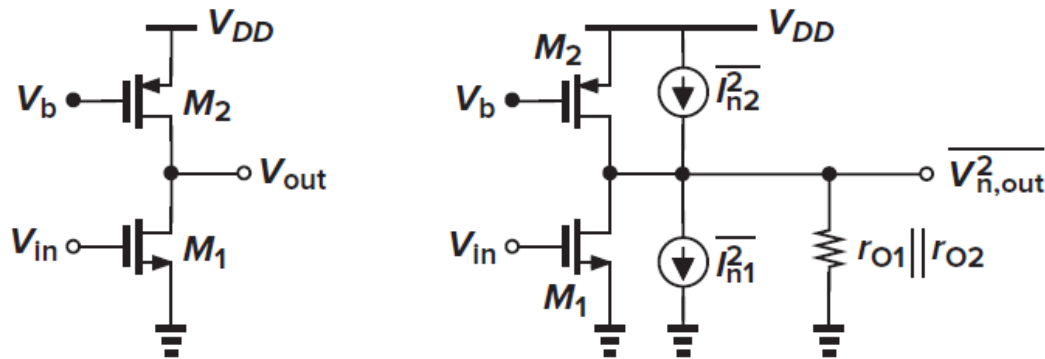
Obtained from above lemma

How to decrease this PSD?



Common-Source Examples

- Common-source stage with current source load

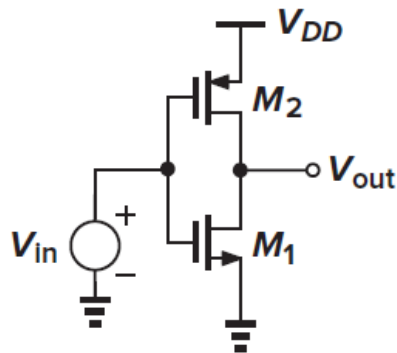


$$\overline{V_{n,in}^2} = 4kT(\gamma g_{m1} + \gamma g_{m2}) \frac{1}{g_{m1}^2}$$

$$= 4kT\gamma \left(\frac{1}{g_{m1}} + \frac{g_{m2}}{g_{m1}^2} \right)$$

Increase g_{m1} and decrease g_{m2} to lower down noise

- Complementary common-source stage

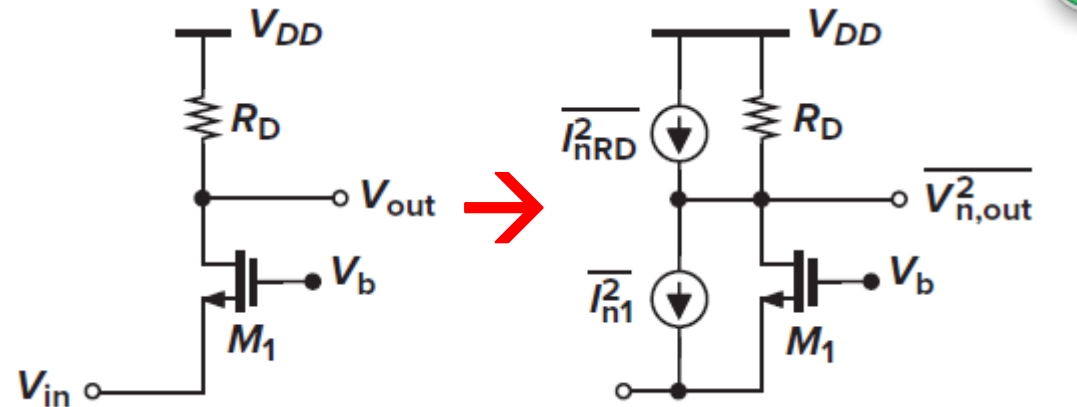


$$\overline{V_{n,in}^2} = \frac{4kT\gamma}{g_{m1} + g_{m2}}$$

Useful low-noise input structure



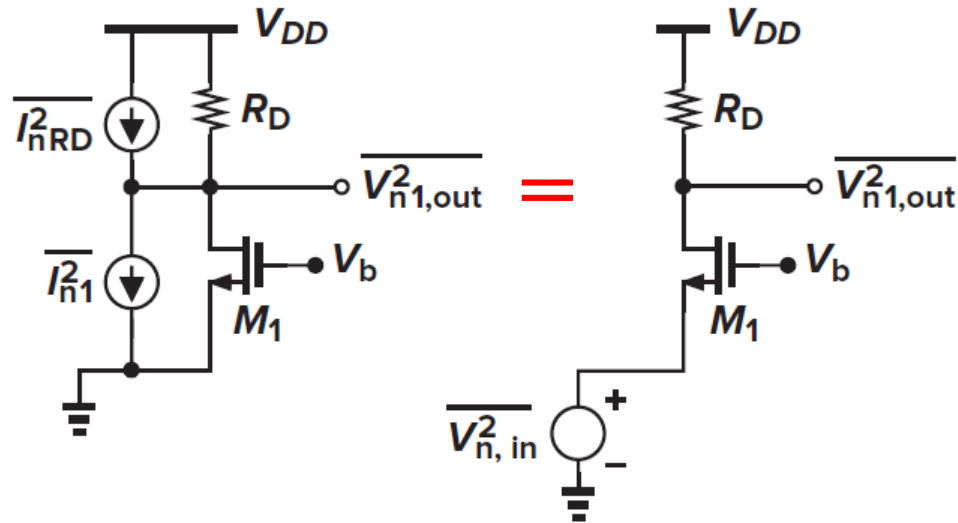
Thermal Noise of CG Stage



- The input impedance is small
 - $I_{n,in}$ can't be neglected even if at low frequencies



Calculating $V_{n,in}$

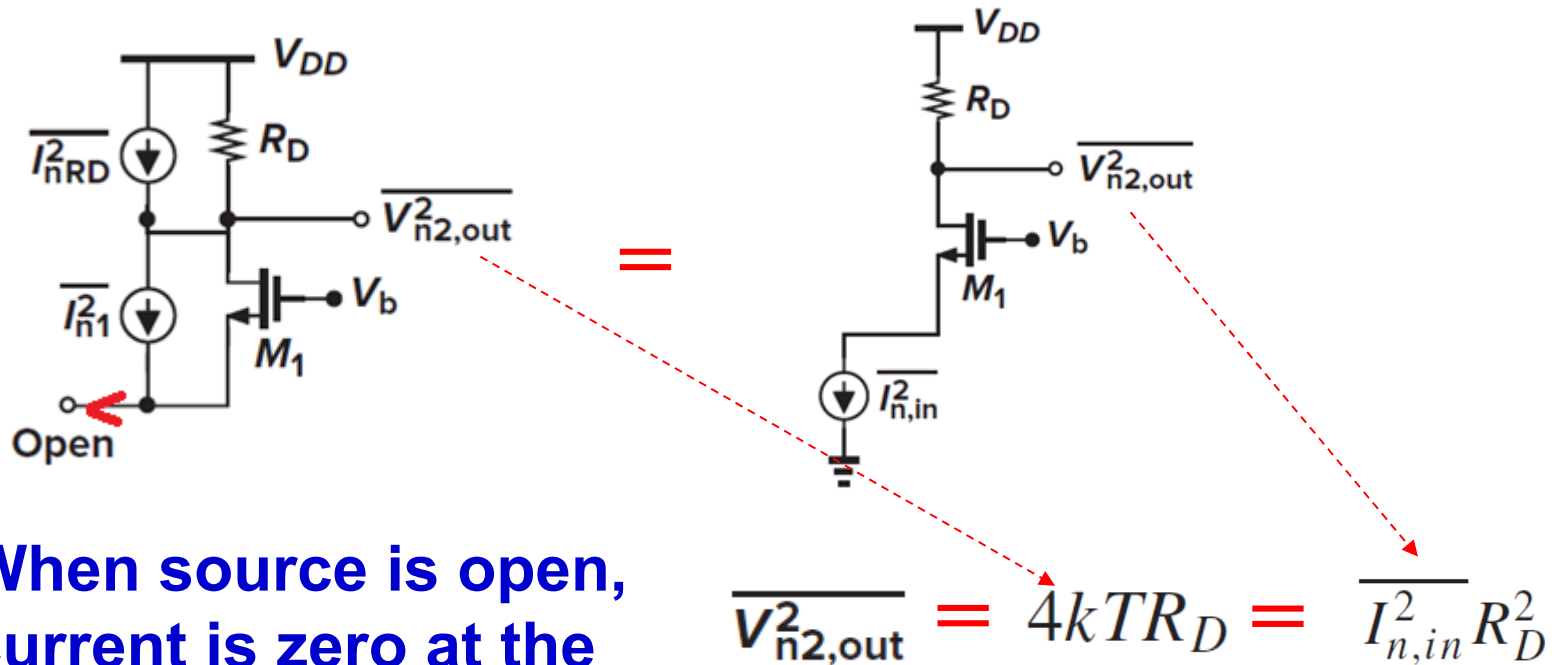


$$\rightarrow \left(4kT\gamma g_m + \frac{4kT}{R_D} \right) R_D^2 = \overline{V_{n,in}^2} (g_m + g_{mb})^2 R_D^2$$

$$\rightarrow \boxed{\overline{V_{n,in}^2} = \frac{4kT(\gamma g_m + 1/R_D)}{(g_m + g_{mb})^2}}$$



Calculating $I_{n,in}$



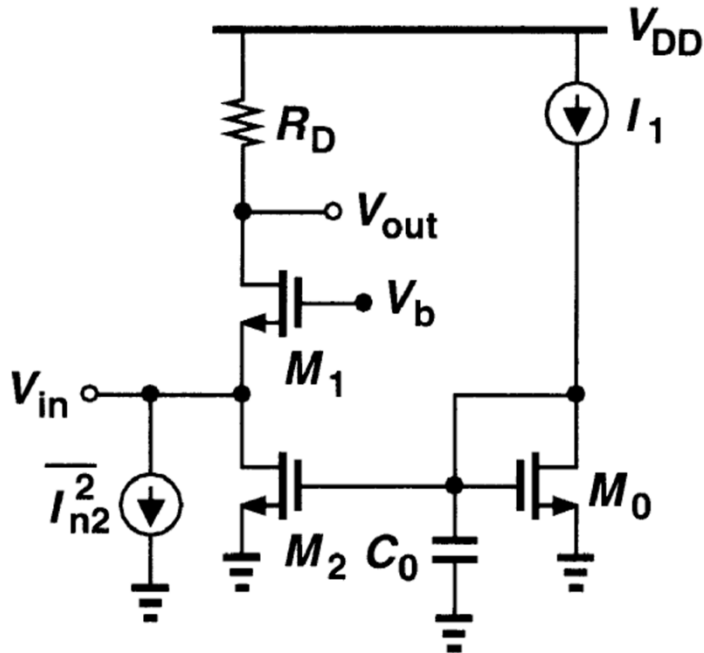
⇒
$$\overline{I_{n,in}^2} = \frac{4kT}{R_D}$$

Drawback of CG stage:

The noise current in the load is directly referred to the input



CG stage with biasing current



- M2 has no contribution to $V_{n,in}$
- But contribute noise current directly to $I_{n,in}$

$$\overline{I_{n2}^2} \approx 4kT\gamma g_{m2}$$

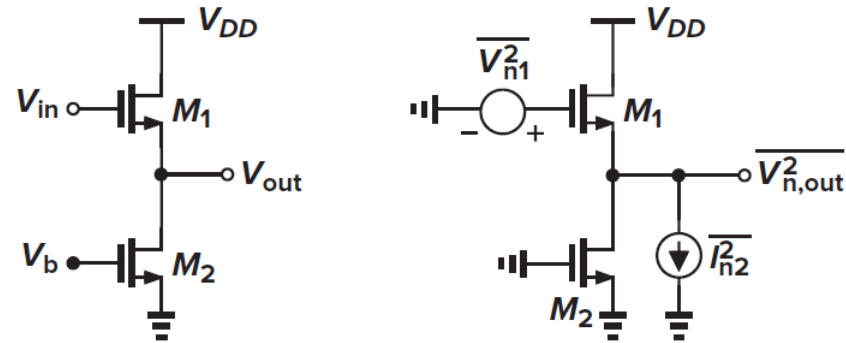
Desire to minimize g_{m2} , but at the price of higher V_{OD} and lower voltage swing ,why?

$$g_{m2} = 2I_{D2}/(V_{GS2} - V_{TH2})$$

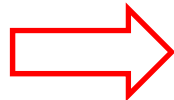


Thermal Noise in Source Follower

- The input impedance is also very large especially at low freq.
 - $I_{n,in}$ of SF stage can often be neglected



$$\overline{V_{n1}^2} = \frac{4kT\gamma}{g_{m1}}, \quad \overline{I_{n2}^2} = 4kT\gamma g_{m2}$$

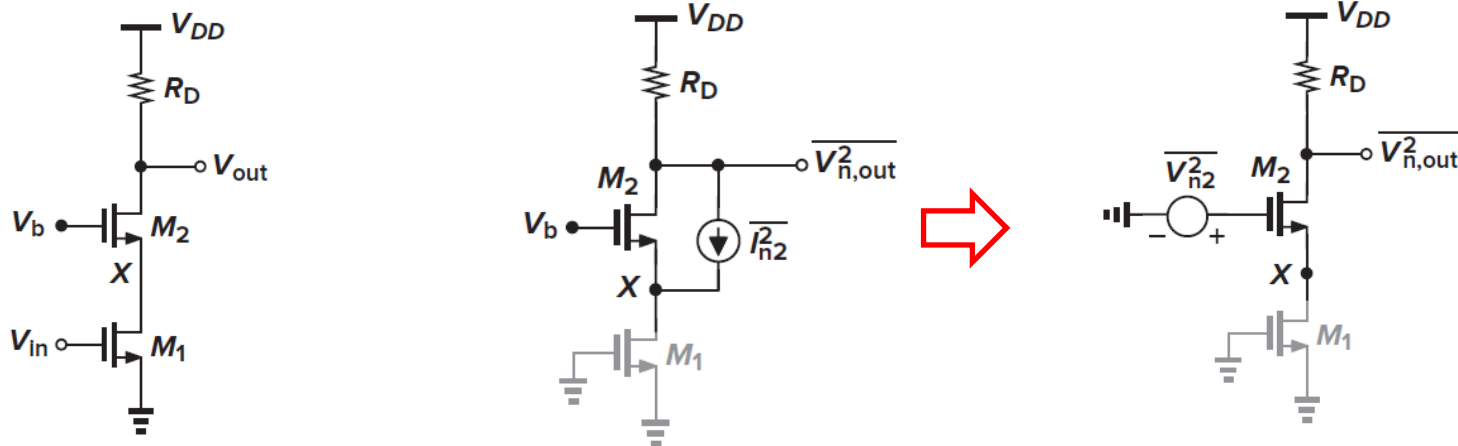


$$\overline{V_{n,in}^2} = 4kT\gamma \left(\frac{1}{g_{m1}} + \frac{g_{m2}}{g_{m1}^2} \right)$$



Thermal Noise in Cascode Stage

- Again, the input impedance is very large especially at low frequency.
 - $I_{n,in}$ of cascode stage can often be neglected



Thermal noise of M1 and R_D

$$\overline{V_{n,in}^2}|_{M1,RD} = 4kT \left(\frac{\gamma}{g_{m1}} + \frac{1}{g_{m1}^2 R_D} \right)$$

How about M_2 ?

M_2 's contribution to $V_{n,out}$ is very small, why?

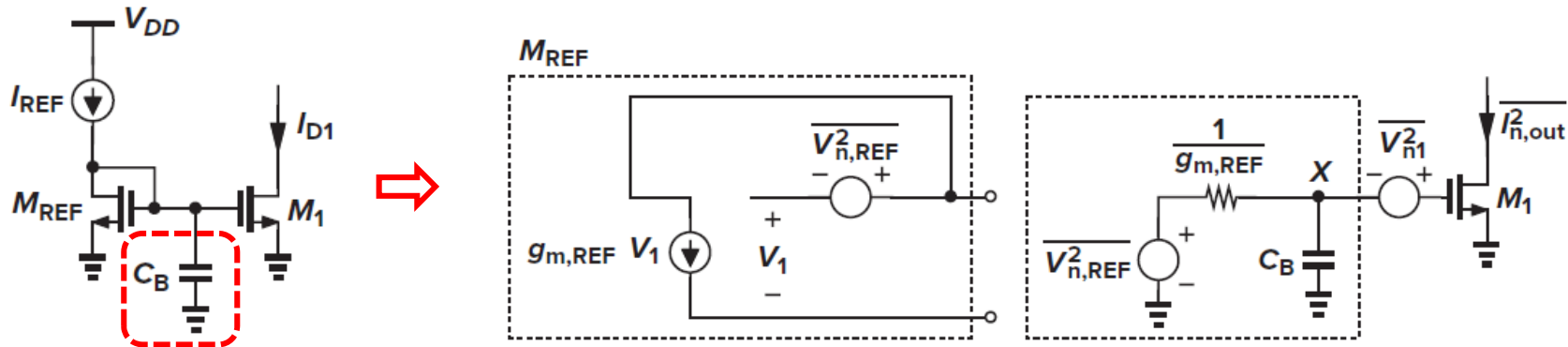


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Be careful about the flicker noise



**Very important
bypass capacitor**

$$\overline{I_{n,out}^2} = \left(\frac{g_{m,REF}^2}{C_B^2 \omega^2 + g_{m,REF}^2} \overline{V_{n,REF}^2} + \overline{V_{n1}^2} \right) g_{m1}^2$$

$$(W/L)_{REF}:(W/L)_1 = 1:N \quad \Rightarrow \quad \overline{I_{n,out}^2} = \left(\frac{N g_{m,REF}^2}{C_B^2 \omega^2 + g_{m,REF}^2} + 1 \right) g_{m1}^2 \overline{V_{n1}^2}$$

To suppress M_{REF} 's noise $\Rightarrow (N-1)g_{m,REF}^2 \ll C_B^2 \omega^2$

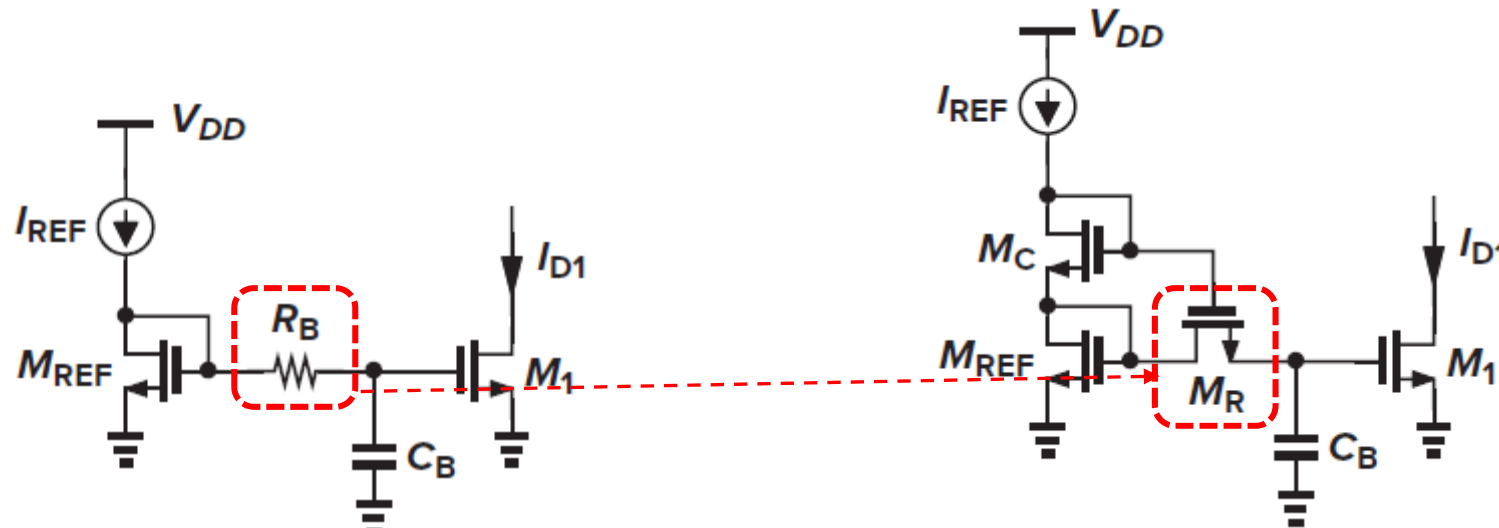
$$\Rightarrow C_B^2 \gg \frac{(N-1)g_{m,REF}^2}{\omega^2}$$



Avoiding very large C_B

- Adding a resistor before C_B

Replacing R_B with M_R to reduce area.



$$\overline{I_{n,out}^2} = \left[\frac{g_{m,REF}^2}{(1 + g_{m,REF} R_B)^2 C_B^2 \omega^2 + g_{m,REF}^2} (\overline{V_{n,REF}^2} + \overline{V_{n,RB}^2}) + \overline{V_{n1}^2} \right] g_{m1}^2$$



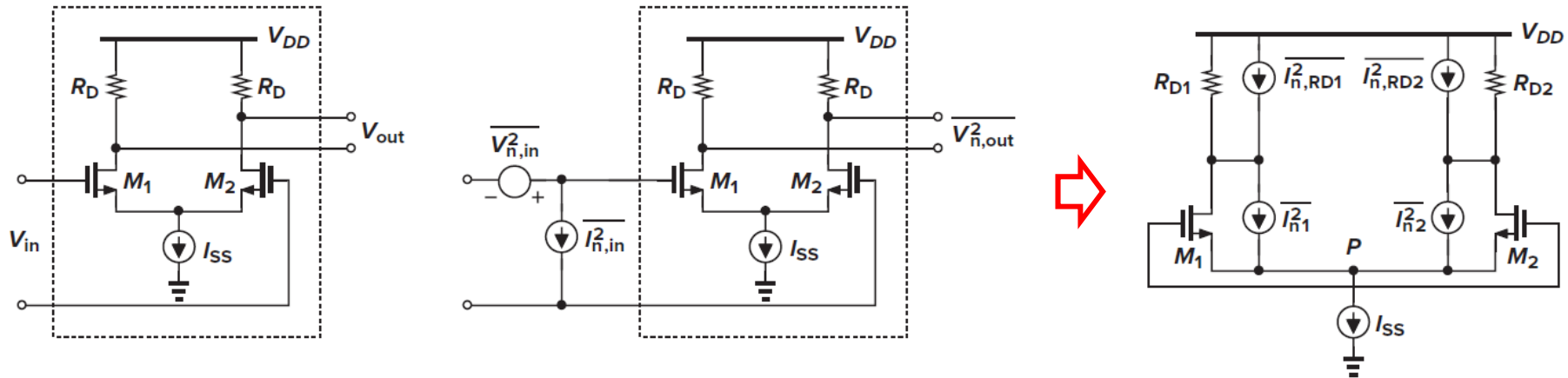
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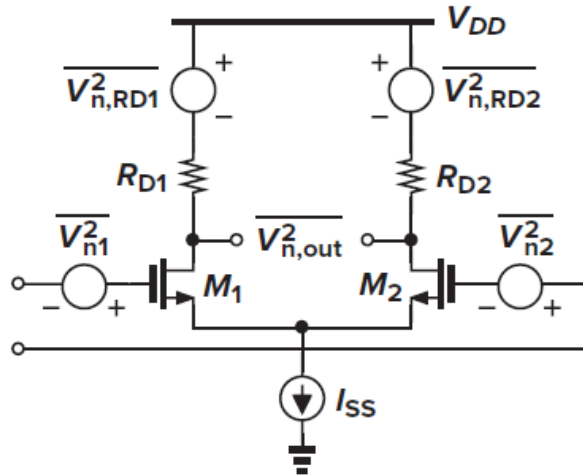


Noise in Differential Pairs

- Modeled as a two-port network
 - $I_{n,in}$ of cascode stage can also be neglected, why?
 - Output noise can be calculated using superposition with input shorted, a little complex



Utilizing the lemma



- Thermal noise of the Input transistors and load resistors

$$\overline{V_{n,in}^2} = 8kT \left(\frac{\gamma}{g_m} + \frac{1}{g_m^2 R_D} \right)$$

- The noise power is doubled (noise voltage increases by a factor of $\sqrt{2}$) compared with a single-ended CS stage

$$\overline{V_{n,in,tot}^2} = 8kT \left(\frac{\gamma}{g_m} + \frac{1}{g_m^2 R_D} \right) + \frac{2K}{C_{ox} W L} \frac{1}{f}$$

- Power consumption is also doubled



The tail current source's noise

- When the differential signal is zero
 - Only contribute common-mode noise
 - No effect to the differential-mode signal
- When there is a small input signal

$$\Delta I_{D1} - \Delta I_{D2} = g_m \Delta V_{in}$$

$$= \sqrt{2\mu_n C_{ox} \frac{W}{L} \left(\frac{I_{SS} + I_n}{2} \right)} \Delta V_{in}$$

I_n is the small noise current

$$\Delta I_{D1} - \Delta I_{D2} \approx \sqrt{2\mu_n C_{ox} \frac{W}{L} \cdot \frac{I_{SS}}{2} \left(1 + \frac{I_n}{2I_{SS}} \right)} \Delta V_{in}$$



$$= g_{m0} \sqrt{\left(1 + \frac{I_n}{2I_{SS}} \right)} \Delta V_{in}$$

Still has effect on the output
Often negligible

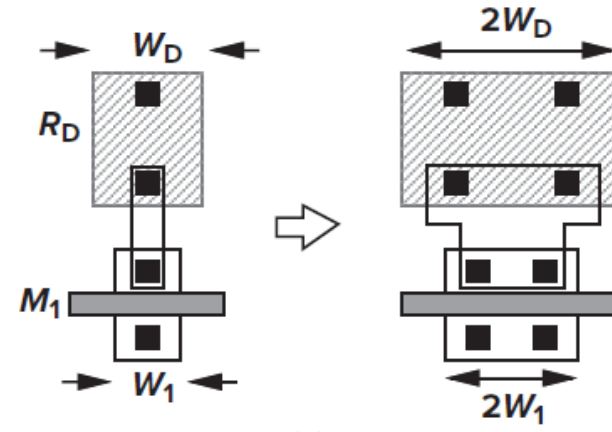
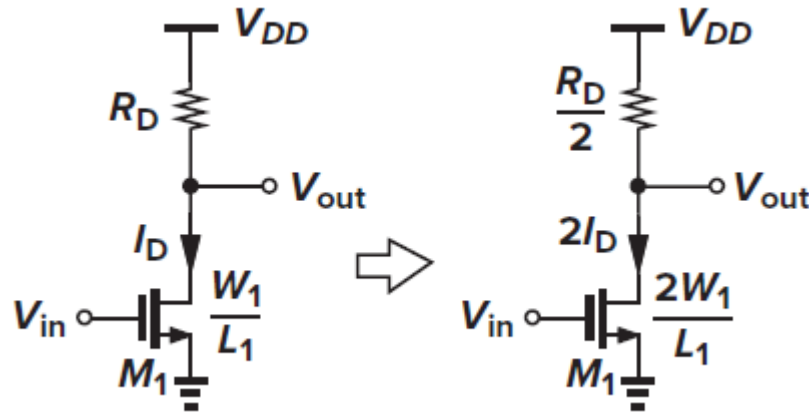


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Noise-Power Trade-Off



$$\overline{V_{n,in}^2} = 4kT \left(\frac{\gamma}{g_m} + \frac{1}{g_m^2 R_D} \right) + \frac{K}{C_{ox} WL} \frac{1}{f} \quad \Rightarrow \quad \overline{V_{n,in}^2} = 4kT \left(\frac{\gamma}{2g_m} + \frac{1}{(2g_m)^2 R_D / 2} \right) + \frac{K}{C_{ox} 2WL} \frac{1}{f}$$

- Noise voltage power is **halved** (voltage decreased by a factor of $\sqrt{2}$)
 - At the price of double power and double area!!
 - This rule holds for many circuits.

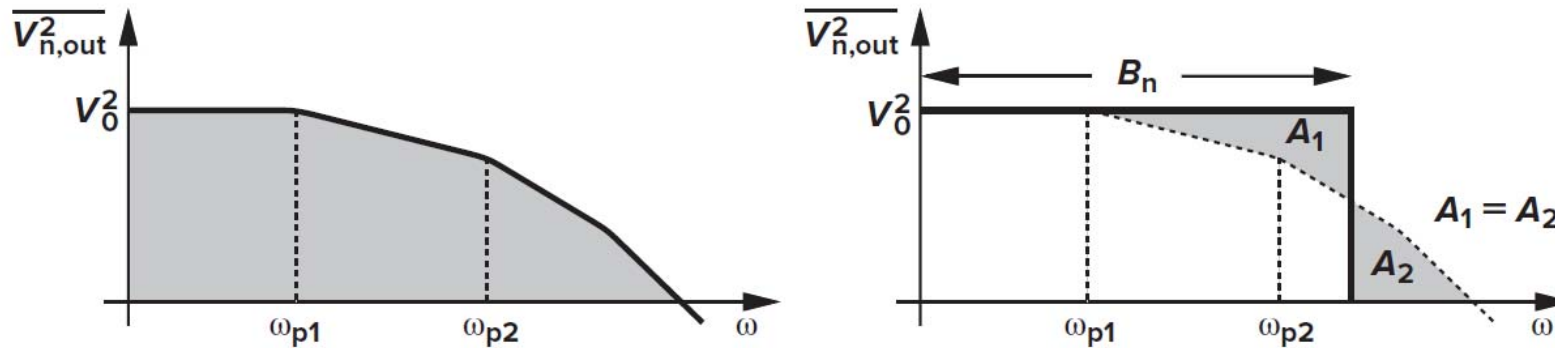


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Noise Bandwidth



$$\overline{V_{n,out,tot}^2} = \int_0^\infty \overline{V_{n,out}^2} df = V_0^2 \cdot B_n$$

- B_n is defined as noise bandwidth
 - For a first-order low-pass system

$$B_n = \frac{\pi}{2} \frac{\omega_0}{2\pi} = \frac{\pi}{2} f_{-3dB}$$



Outline

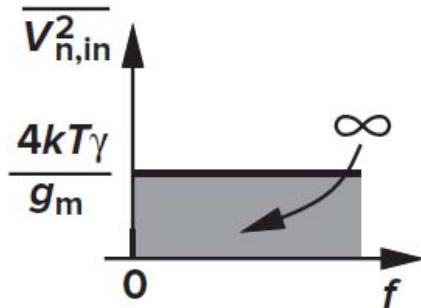
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Problem of noise integration

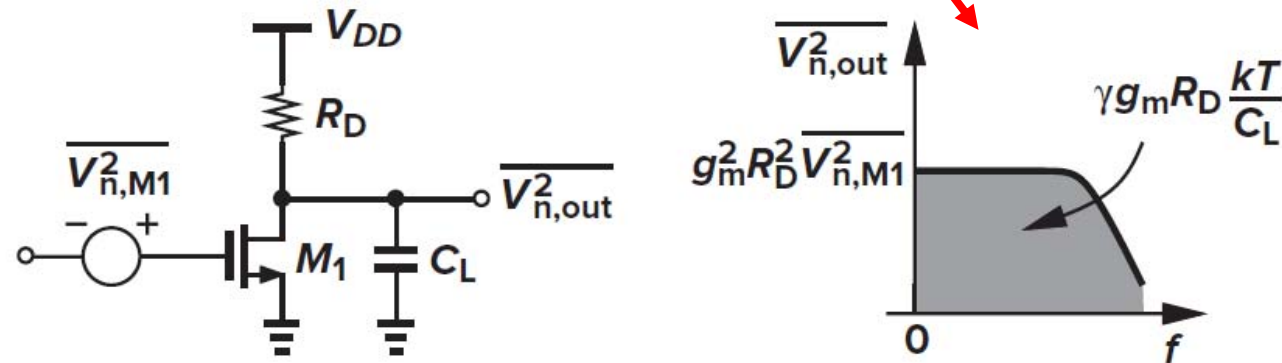
Integration at the input

Incorrect!



Integration at the output

Correct



Solution:

(1) Integration at the output, then divide the result by the low-frequency gain;

(2) Input-referred PSD is obtained as the output PSD divided by the low-frequency gain, then integrated at the input



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 - **CS**
 - CG
 - SF
 - **Cascode**
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